EA ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 January 2026 Worked Solutions

Jan 2026 paper rebuilt with the worked solution after each question.

9
paper questions

WMA11
Pure 1

Single-paper solutions
standalone IAL route

Dr Eslam Ahmed

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 January 2026

Jan 2026 | 9 questions | 75 marks

9
questions

75
marks

Answers
worked solution after each question

Question 1

6 marks

Integration

1.

$$y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \qquad x > 0$$

Find, in simplest form,

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x}$

(3)

(b) $\displaystyle\int y \,\mathrm{d}x$

(3)

(Total for Question 1 is 6 marks)

WMA11/01 JANUARY 2026

Worked Solution - Question 1

Integration

Topic group

1. Rewrite with powers

$$y = 4x^3 + \frac{3}{2}x^{-1/2} - 8.$$

2. Differentiate

$$\frac{dy}{dx} = 12x^2 + \frac{3}{2}\left(-\frac{1}{2}\right)x^{-3/2}.$$

3. Simplify derivative

$$\frac{dy}{dx} = 12x^2 - \frac{3}{4}x^{-3/2} = 12x^2 - \frac{3}{4x^{3/2}}.$$

4. Integrate

$$\int y\,dx = \int \left(4x^3 + \frac{3}{2}x^{-1/2} - 8\right)dx.$$

5. Simplify integral

$$\int y\,dx = x^4 + 3x^{1/2} - 8x + c = x^4 + 3\sqrt{x} - 8x + c.$$

Final answer

(a) $\dfrac{dy}{dx} = 12x^2 - \dfrac{3}{4x^{3/2}}.$

(b) $\int y\,dx = x^4 + 3\sqrt{x} - 8x + c.$

Question 2

6 marks

Simultaneous Equations

2. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

(a) Given

$$y - x = 4$$

and

$$2x^2 - xy = 8$$

show that

$$x^2 - 4x - 8 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$y - x = 4$$
$$2x^2 - xy = 8$$

writing the answers in the form $p + q\sqrt{3}$, where p and q are integers to be found.

(4)

(Total for Question 2 is 6 marks)

Worked Solution - Question 2

Simultaneous Equations

Topic group

1. Use the linear equation

$y - x = 4$, so $y = x + 4$.

2. Substitute into the second equation

$2x^2 - xy = 8$ becomes $2x^2 - x(x+4) = 8$.

3. Show the quadratic

$2x^2 - x^2 - 4x = 8$, so $x^2 - 4x - 8 = 0$.

4. Complete the square

$x^2 - 4x - 8 = 0$ gives $(x-2)^2 = 12$.

5. Find x

$x = 2 \pm \sqrt{12} = 2 \pm 2\sqrt{3}$.

6. Find y

Since $y = x + 4$, $y = 6 \pm 2\sqrt{3}$ with matching signs.

7. State both solutions

$(x, y) = (2 + 2\sqrt{3}, 6 + 2\sqrt{3})$ or $(2 - 2\sqrt{3}, 6 - 2\sqrt{3})$.

Final answer

$(x, y) = (2 + 2\sqrt{3},\ 6 + 2\sqrt{3})$ or $(2 - 2\sqrt{3},\ 6 - 2\sqrt{3})$.

WMA11/01 JANUARY 2026

Question 3

5 marks

Inequalities

3. (i) Find the range of values of z for which

$$\frac{4}{z} < 8$$

giving the answer in simplest form.

(2)

(ii)

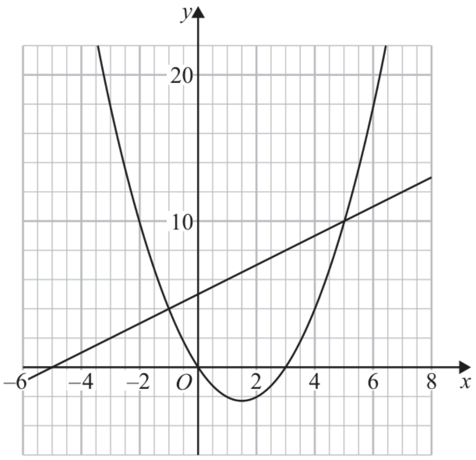


Figure 1

Figure 1 shows a plot of

- the line with equation $y = x + 5$
- the curve with equation $y = x(x - 3)$

On the opposite page there is a copy of Figure 1 labelled Diagram 1.
On Diagram 1 represent the following inequality graphically.

$$x + 5 \leqslant y \leqslant x(x - 3) \quad \text{for} \quad x \geqslant -2$$

(3)

Question 3 continued

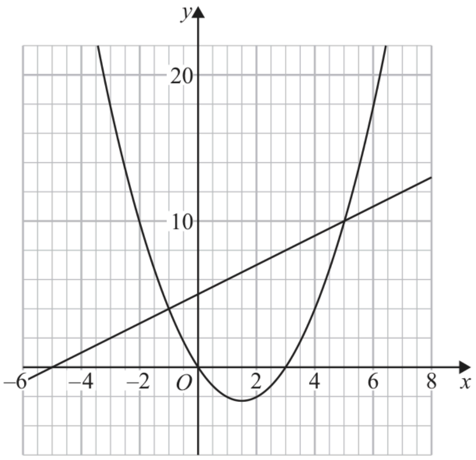


Diagram 1

(Total for Question 3 is 5 marks)

WMA11/01 JANUARY 2026

Worked Solution - Question 3

Inequalities

Topic group

1. Solve the rational inequality

$\frac{4}{z} < 8$ gives $\frac{4}{z} - 8 < 0$, so $\frac{4 - 8z}{z} < 0$.

2. Use critical values

The critical values are $z = 0$ and $z = \frac{1}{2}$.

3. State the range

A sign check gives $z < 0$ or $z > \frac{1}{2}$.

4. Find where line and curve meet

$x + 5 = x(x - 3)$ gives $x^2 - 4x - 5 = 0$, so $(x - 5)(x + 1) = 0$.

5. Combine with x greater than or equal to -2

The inequality $x + 5 \leq y \leq x(x - 3)$ is possible when $x \leq -1$ or $x \geq 5$. With $x \geq -2$, the shaded x-ranges are $-2 \leq x \leq -1$ and $x \geq 5$.

6. Graphical representation

Draw the vertical boundary $x = -2$ and shade the two regions between the line and the parabola: from $x = -2$ to $x = -1$, and to the right of $x = 5$.

Final answer

(i) $z < 0$ or $z > \frac{1}{2}$. (ii) shade the region above $y = x + 5$, below $y = x(x - 3)$, with $x \geq -2$.

Question 4

9 marks

Straight Line

4. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers.
(3)

The line L_2

- passes through A and is perpendicular to L_1
- cuts the x-axis at the point C

(b) Find the coordinates of C.
(3)

(c) Hence find the area of triangle ABC.
(3)

(Total for Question 4 is 9 marks)

Worked Solution - Question 4

Straight Line

Topic group

1. Gradient of L1

Using $A(-1, 8)$ and $B(7, 2)$, $m = \dfrac{2-8}{7-(-1)} = -\dfrac{6}{8} = -\dfrac{3}{4}$.

2. Equation of L1

$y - 2 = -\dfrac{3}{4}(x - 7)$.

3. Rearrange

$4y - 8 = -3x + 21$, hence $3x + 4y - 29 = 0$.

4. Gradient of L2

Since L_2 is perpendicular to L_1, its gradient is $\dfrac{4}{3}$.

5. Equation of L2

L_2 passes through $A(-1, 8)$, so $y - 8 = \dfrac{4}{3}(x + 1)$.

6. Find C

At the x-axis, $y = 0$, so $-8 = \dfrac{4}{3}(x + 1)$. Hence $x + 1 = -6$ and $x = -7$, giving $C = (-7, 0)$.

7. Find perpendicular side lengths

$AB = \sqrt{8^2 + (-6)^2} = 10$ and $AC = \sqrt{(-6)^2 + (-8)^2} = 10$.

8. Area of triangle ABC

Since $AB \perp AC$, $\text{Area} = \dfrac{1}{2}(10)(10) = 50$.

Final answer

$(a)\ 3x + 4y - 29 = 0$.

$(b)\ C = (-7, 0)$.

$(c)\ 50$.

Question 5

9 marks

Radians

5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

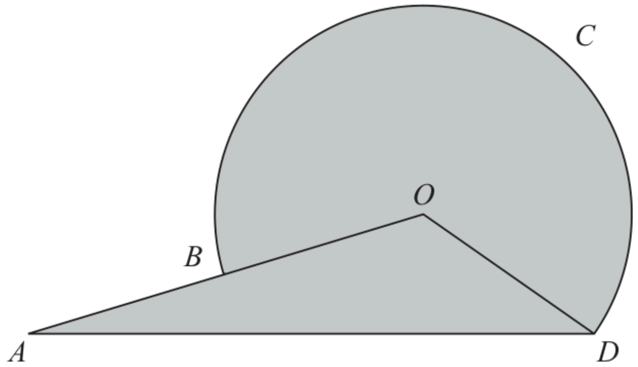


Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector *BODCB* of a circle centre *O* joined to a triangle *AOD*.

Given that

- *ABO* is a straight line
- $OD = 3\,\text{cm}$
- $AD = 8\,\text{cm}$
- angle $AOD = 2.5$ radians

(a) show that angle *ODA* is 0.415 radians to 3 significant figures.

(3)

(b) Find the total area of the logo, in cm^2, to one decimal place.

(3)

(c) Find the perimeter of the logo, *ABCDA*, in cm, to one decimal place.

(3)

(Total for Question 5 is 9 marks)

Worked Solution - Question 5

Radians

Topic group

1. Use triangle AOD

In triangle AOD, $OD = 3$, $AD = 8$ and $\angle AOD = 2.5$ radians.

2. Use the sine rule

$\dfrac{\sin \angle OAD}{3} = \dfrac{\sin 2.5}{8}$, so $\angle OAD = \sin^{-1}\left(\dfrac{3\sin 2.5}{8}\right) = 0.2263\ldots$.

3. Show angle ODA

$\angle ODA = \pi - 2.5 - 0.2263\ldots = 0.4152\ldots$, so $\angle ODA = 0.415$ radians to 3 significant figures.

4. Area of the major sector

The sector has radius 3 and angle $2\pi - 2.5$, so its area is $\dfrac{1}{2}(3)^2(2\pi - 2.5) = 17.02\ldots$

5. Area of triangle AOD

$\text{Area of triangle } AOD = \dfrac{1}{2}(3)(8)\sin 0.4152\ldots = 4.84\ldots$

6. Total area

$17.02\ldots + 4.84\ldots = 21.86\ldots$, so the area is $21.9\ \mathrm{cm}^2$ to one decimal place.

7. Find AO

By the cosine rule, $AO^2 = 3^2 + 8^2 - 2(3)(8)\cos 0.4152\ldots$, so $AO \approx 5.39$ cm.

8. Find AB

Since A, B, O are collinear and $OB = 3$, $AB = AO - OB \approx 5.39 - 3 = 2.39$ cm.

9. Perimeter

$AB + \text{major arc } BD + AD = 2.39\ldots + 3(2\pi - 2.5) + 8 = 21.7 \text{ cm}$ to one decimal place.

Final answer

(a) $\angle ODA = 0.415$ rad.

(b) 21.9 cm^2.

(c) 21.7 cm.

WMA11/01 JANUARY 2026

Question 6

9 marks

Polynomials

6.

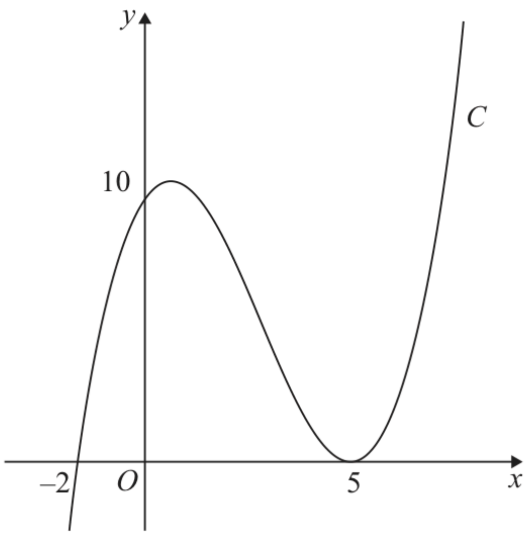


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a cubic function in x.

The curve C

- cuts the x-axis at $(-2, 0)$ and cuts the y-axis at $(0, 10)$
- touches the x-axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $\mathrm{f}\left(\frac{1}{3}x\right) = 0$

(ii) $\mathrm{f}(x - 3) = 0$

(2)

(b) Find an expression for $\mathrm{f}(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(*Solutions based entirely on calculator technology are not acceptable.*)

(4)

(Total for Question 6 is 9 marks)

WMA11/01 JANUARY 2026

Worked Solution - Question 6

Polynomials

Topic group

1. Roots of f one third x

The roots of $f(x) = 0$ are $x = -2$ and $x = 5$, with $x = 5$ repeated. For $f\left(\frac{1}{3}x\right) = 0$, set $\frac{1}{3}x = -2$ or 5, giving $x = -6$ or 15.

2. Roots of f of x minus 3

For $f(x-3) = 0$, set $x - 3 = -2$ or 5, giving $x = 1$ or 8.

3. Use the roots and repeated root

Since the curve cuts at $(-2, 0)$ and touches at $(5, 0)$, write $f(x) = k(x+2)(x-5)^2$.

4. Use the y-intercept

The curve passes through $(0, 10)$, so $10 = k(2)(25) = 50k$.

5. State f(x)

$k = \frac{1}{5}$, hence $f(x) = \frac{1}{5}(x+2)(x-5)^2$.

6. Set equal to the line

$\frac{1}{5}(x+2)(x-5)^2 = 10(x+2)$.

7. First solution

The factor $x + 2$ gives one solution $x = -2$.

8. Find the other solutions

For $x \neq -2$, divide by $x + 2$: $\frac{1}{5}(x-5)^2 = 10$, so $(x-5)^2 = 50$.

9. State all intersections

$x = 5 \pm \sqrt{50} = 5 \pm 5\sqrt{2}$. Therefore the three x-coordinates are -2, $5 - 5\sqrt{2}$ and $5 + 5\sqrt{2}$.

Final answer

$(a)(i)\ -6, 15. \quad (a)(ii)\ 1, 8.$

$(b)\ f(x) = \dfrac{1}{5}(x+2)(x-5)^2.$

$(c)\ x = -2,\ 5 \pm 5\sqrt{2}.$

Question 7

10 marks

Differentiation

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{\mathrm{d}y}{\mathrm{d}x}$ can be written in the form

$$p(x+q)^2 + r$$

where p, q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{\mathrm{d}y}{\mathrm{d}x}$

(ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,

(c) find the equation of the tangent to C at S, giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

WMA11/01 JANUARY 2026

Worked Solution - Question 7

Differentiation

Topic group

1. Differentiate

$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$ gives $\frac{dy}{dx} = 2x^2 - 16x + 43$.

2. Complete the square

$2x^2 - 16x + 43 = 2(x^2 - 8x) + 43 = 2[(x-4)^2 - 16] + 43$.

3. Simplify

$\frac{dy}{dx} = 2(x-4)^2 + 11$.

4. Minimum gradient

Since $2(x-4)^2 \geq 0$, the minimum value of $\frac{dy}{dx}$ is 11.

5. Where it occurs

The minimum occurs when $x - 4 = 0$, so $x = 4$.

6. Find the point S

At $x = 4$, $y = \frac{2}{3}(4)^3 - 8(4)^2 + 43(4) - \frac{20}{3} = 80$.

7. Tangent equation

The gradient at S is 11, so $y - 80 = 11(x - 4)$.

8. Simplify

$y = 11x + 36$.

Final answer

$(a)\ \dfrac{dy}{dx} = 2(x-4)^2 + 11.$

$(b)\ \min \dfrac{dy}{dx} = 11 \text{ at } x = 4.$

$(c)\ y = 11x + 36.$

Question 8

12 marks

Integration

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$, $\quad x > 0$

A point P lies on C.

Given that

- $\mathrm{f}'(x) = 4x^2 + \dfrac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P,

(ii) find, using algebra, the other possible x coordinate of P.

(6)

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P.

(2)

(c) Find $\mathrm{f}(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

WMA11/01 JANUARY 2026

Worked Solution - Question 8

Integration

Topic group

1. Verify x equals root 3

$f'(\sqrt{3}) = 4(\sqrt{3})^2 + \dfrac{6}{(\sqrt{3})^2} - 4 = 12 + 2 - 4 = 10$, which matches the tangent gradient.

2. Find all possible x-coordinates

Set $4x^2 + \dfrac{6}{x^2} - 4 = 10$.

3. Form a quadratic in x squared

Multiplying by x^2 gives $4x^4 - 14x^2 + 6 = 0$, so $2x^4 - 7x^2 + 3 = 0$.

4. Factor

$2x^4 - 7x^2 + 3 = (2x^2 - 1)(x^2 - 3) = 0$.

5. Use x greater than zero

$x = \sqrt{3}$ or $x = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$. The other possible x-coordinate is $\dfrac{\sqrt{2}}{2}$.

6. Find the point P

Given $x_P = \sqrt{3}$, the tangent gives $y_P = 10\sqrt{3} - 6\sqrt{3} = 4\sqrt{3}$.

7. Normal equation

The normal gradient is $-\dfrac{1}{10}$, so $y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3})$.

8. Integrate f prime

$f(x) = \int \left(4x^2 + \dfrac{6}{x^2} - 4\right) dx = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + c$.

9. Use P to find c

$$4\sqrt{3} = \frac{4}{3}(\sqrt{3})^3 - \frac{6}{\sqrt{3}} - 4\sqrt{3} + c = 4\sqrt{3} - 2\sqrt{3} - 4\sqrt{3} + c.$$

10. State f(x)

$c = 6\sqrt{3}$, so $f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}$.

Final answer

$(a)(i)$ verified.

$(a)(ii)$ $x = \dfrac{\sqrt{2}}{2}$.

(b) $y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3})$.

(c) $f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}$.

WMA11/01 JANUARY 2026

Question 9

9 marks

Trigonometric Functions

9.

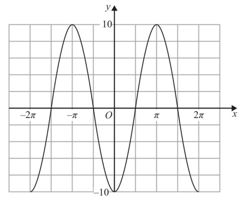


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = \mathrm{f}(x)$.

(a) Write down an expression for $\mathrm{f}(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$\mathrm{f}(x) = 10 - x^2 \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$\mathrm{f}(x) = \tan x \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

giving a reason for your answer.

(4)

(a) ....

(b)(i)

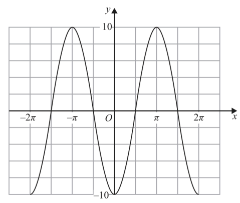
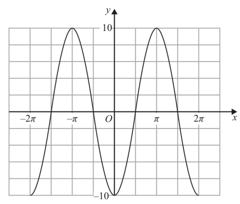


Diagram 1



Copy of Diagram 1

Only use this copy if you need to redraw your graph.

(b)(ii) ....

(c)(i)

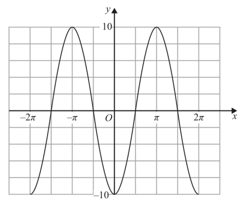
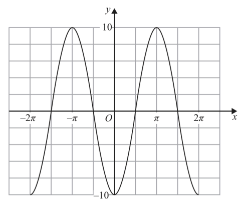


Diagram 2



Copy of Diagram 2

Only use this copy if you need to redraw your graph.

(c)(ii) ....

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

Trigonometric Functions

Topic group

1. Identify f(x)

The graph has amplitude 10, period 2π, and value -10 at $x = 0$, so $f(x) = -10\cos x$.

2. Sketch y equals 10 minus x squared

The graph $y = 10 - x^2$ is a downward-opening parabola with vertex $(0, 10)$ and x-intercepts at $x = \pm\sqrt{10}$.

3. Count intersections with the parabola

On the given sketch, the parabola meets $f(x)$ twice. Outside the central part the parabola is below the range of $f(x)$, so there are 2 solutions in $-100\pi \leq x \leq 100\pi$.

4. Sketch y equals tan x

The graph $y = \tan x$ has zeros at integer multiples of π and vertical asymptotes at $x = \dfrac{\pi}{2} + n\pi$.

5. Count in the reference interval

In $-2\pi \leq x \leq 2\pi$, the graph $y = \tan x$ intersects $f(x)$ four times.

6. Scale to the required interval

The interval $-100\pi \leq x \leq 100\pi$ is 50 times as long as $-2\pi \leq x \leq 2\pi$, so the number of solutions is $50 \times 4 = 200$.

Final answer

(a) $f(x) = -10\cos x$.

$(b)(ii)$ 2.

$(c)(ii)$ 200.